INTERIM FEDERAL SPECIFICATION

~~VALVE, FLUSH;~~ BULB, FLUSH; GUIDE ARM; LIFT RODS; AND TRIP LEVER ASSEMBLY; (WATER CLOSET TANK) (LAND USE)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of flush valve for use in water closet tanks. It also covers the flush valve bulb, lift rods, guide arm, and trip lever assembly for use with the flush valve.

1.1.1 Specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available. This specification is intended to cover the items, indicated by the title, generally used by the Federal Government.

1.2 Classification.

1.2.1 Definitive part number (DPN). Items furnished in accordance with this specification shall be identified by a DPN. This number is intended for cataloging and ordering purposes; it is not intended for surface printing on items in accordance with this specification. The DPN shall consist of the Federal Specification symbol plus the specification part number (SPN) (see 1.2.2). The DPN shall be written as indicated below (see 6.2 and 6.3):

DPN ---------- WWV1916 - XX
Fed. Spec. symbol ----------
SPN (see 1.2.2) ----------

1.2.2 SPN. The SPN shall be a two position numeric code which identifies items in accordance with this specification (see table I).

TABLE I. SPN identification

SPN	Item identification	Paragraph reference
01	Flush valve	3.4
02	Guide arm	3.4.2.e
03	Actuating rod, 9-inch (upper lift wire)	3.4.2.f
04	Lift rod, 4-1/2-inch (lower lift wire)	3.4.2.g
05	Lift rod, 5-1/2-inch (lower lift wire)	3.4.2.g
06	Flush valve, bulb	3.5
07	Trip lever assembly	3.6

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-C-320 - Chromium Plating (Electrodeposited).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Setup.

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Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D570 - Test for Water Absorption of Plastics.

(Application for copies shall be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard product. Except as otherwise specified herein, the flush valves and trip lever assemblies shall be the manufacturer's current standard model. Items in accordance with this specification shall be furnished new and unused.

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3.2 Interchangeability. Identical items furnished under a specific contract shall permit interchangeability of replaceable parts and assemblies.

3.3 Material. Material shall conform to requirements specified hereinafter. Material not specified shall be of the type and quality recognized by industry for the intended purpose.

3.3.1 Plastics. Plastic parts continually immersed in water shall not absorb water in amounts greater than 0.42 percent by weight when tested in accordance with 4.5.1.

3.4 Flush valve (SPN - 01).

3.4.1 Component parts. The flush valve shall consist of the following component parts:

a. Flush valve base.
b. Flush valve gasket.
c. Flush valve lock nut.
d. Overflow tube.
e. Guide arm (SPN - 02).
f. Actuating rod (upper lift wire, SPN - 03).
g. Lift rod (lower lift wire, SPN - 04).

3.4.2 Construction. The component parts of the flush valve shall be constructed in accordance with the requirements specified herein. The parts shall be sufficiently rigid and of adequate strength to maintain utility and to prevent distortion during operation. All threads shall be in accordance with H28.

a. Flush valve base. The flush valve base shall be of cast brass or other suitable non-corrosive casting alloy. The base dimensions shall be in accordance with figure 1. The shape, plane, and dimensions of the valve seat shall be such as to seat a flush valve bulb in accordance with 3.5. The base overflow tube receiver shall be threaded with 1-27 UNS-2B threads. The minimum number of full threads in the overflow tube receiver shall be four.

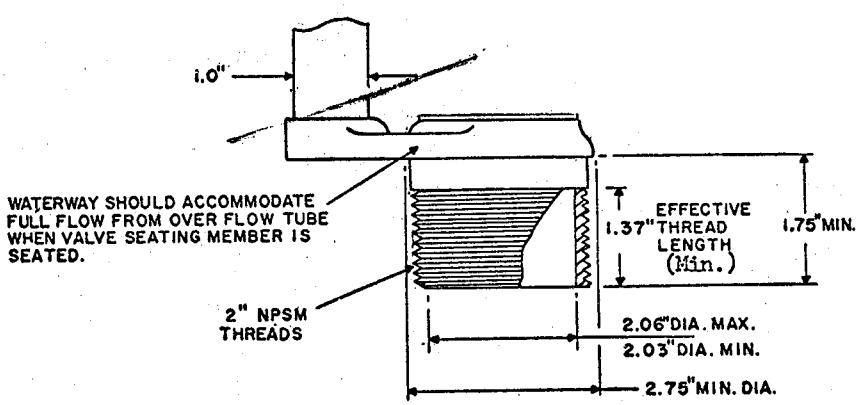


Figure 1. FLUSH VALVE BASE

b. Flush valve gasket. Flush valve gaskets shall be of an elastometric material. The gasket shall form a seal to prevent leaks and also serve as a cushion to prevent cracking tank when the lock nut is tightened.

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c. Flush valve lock nut. The lock nut shall be of cast brass or suitable copper casting alloy. The nut shall be threaded with 2 - 11-1/2 NPSM internal threads. The minimum dimensions, in inches, of the lock nut shall be as follows:

Outside diameter	3.00
Overall thickness	0.34
Width between flats	2.50
Height of flat	0.25

d. Overflow tube. The overflow tube shall be of seamless or welded brass or hard drawn copper tubing having a minimum wall thickness of 0.030 inch. The nominal outside diameter shall be 1 inch. The length of the tube shall not be less than 11 inches. The tube shall be threaded at one end with 1-27 UNS - 2A threads for a length of 1/4 ± 1/16 inch.

e. Guide arm (SPN - 02). The flush bulb guide arm shall be cast brass. The clamp shall be adjustable and shall clamp securely to a 1 inch O.D. overflow tube by means of thumb screws. The adjustment range shall be adequate to support the lift rod at any position between 1-1/2 and 2 inches from the nearest surface of the tube. The guide adjustment shall be secured by means of a slotted-head screw or thumb screw. The guide rod hole shall have a diameter between 9/64- and 5/32-inch, and a length of not less than 1/2-inch.

f. Actuating rod (upper lift wire, SPN - 03). The actuating rod shall be of brass or hard drawn copper wire. The diameter of the rod shall be 0.114 ± 0.001-inch (American Wire Gage (AWG) No. 9. The rod shall be bent and looped at one end. The angle of the bend shall be 90 degrees. The inside diameter of the loop shall be 3/16 to 1/4-inch. The plane of the loop shall be perpendicular to the longitudinal axis of the rod. The length from the bend to the outside of the loop shall range from 1/2 to 3/4-inch. The length of the rod from the bend shall be 9 ± 1/8-inches.

g. Lift rod (lower lift wire, SPN's 04 and 05). The lift rod shall be of the materials described in "f" above. The rod shall be looped at one end. The outside diameter of the loop shall be 3/8 ± 1/16-inch. The straight end shall be threaded with .138-32 UNC - 2A threads. The thread length shall be not less than 3/8-inch. The overall length of the SPN - 04 rod shall be 4-1/2 ± 1/8-inch. The length of the SPN - 05 rod shall be 5-1/2 ± 1/8-inches (see 1.2.2).

3.5 Flush bulb (SPN - 06). The flush bulb shall be a one piece semi-spherical rubber-like ball. The diameter of the bulb shall be 2-1/2 ± 1/8-inches. The height of the bulb shall be 2-3/8 ± 1/8-inches. The top of the bulb shall be reinforced. The bulb shall have a threaded copper alloy insert at the top to receive the lift rod. The insert shall be threaded with .138-32 UNC - 2B threads. The thread length shall be not less than 3/8-inch. The bulb shall be designed to seat on a base in accordance with 3.4.2.a.

3.6 Trip lever assembly (SPN - 07). The trip lever assembly shall be single action and designed for front mounting. The assembly shall consist essentially of a hanger or frame having a threaded mounted bushing, a lift arm and a handle to actuate the lift arm. The hanger and bushing shall be furnished complete with lock nut, rubber cushion, washer, and all components required for assembly. The hanger and bushing shall fit the standard opening of a vitreous china tank. The lift arm shall be mounted directly on the handle shaft or shall be integral with the shaft. The lift arm shall be provided with at least three 3/16-inch minimum diameter holes to attach the flush bulb actuating rod. The free end of the arm shall be located at a minimum distance of 8-inches from the center of the handle shaft, and offset approximately 2 to 3-inches from the front of the tank. Adequate stops shall be provided to limit lift arm travel. The operating handle shall have a minimum length of 2-1/2-inches when measured from the center of the handle shaft. The handle shall operate when motion is downward. All metal parts of the assembly, except the operating handle, shall be copper-zinc alloy. The handle shall be bright chrome plated in accordance with QQ-C-320, class 1, type I.

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3.7 Workmanship. Workmanship shall be in accordance with the requirements specified herein and shall meet the quality of workmanship customarily applied in the construction of water closet plumbing fittings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Lot. A lot shall consist of all items with the same DPN offered for delivery at one time.

4.2.2 Sampling for examination and test. Select sample size for examination in accordance with level S-4 in MIL-STD-105. Select sample size for test in accordance with level S-1 in MIL-STD-105.

4.3 Examination. Examine items as applicable for defects listed in table II. The acceptable quality levels (AQL's) shall be 2.5 percent defective for major defects and 6.5 percent defective for total defects.

TABLE II. Classification of defects, end item

Defects	Major	Minor
Specification part number not as specified	X	
Parts not interchangeable (see 3.2)	X	
Materials not as specified	X	
Flush valve parts missing (see 3.4)	X	
Threads not as specified		X
Dimensions not as specified	X	
Guide arm not as specified (see 3.4.2.e)	X	
Upper lift wire not as specified (see 3.4.2.f)	X	
Lower lift wire not as specified (see 3.4.2.g)	X	
Flush valve bulb not semi-spherical (see 3.5)		X
Trip lever handle not plated (see 3.6)	X	
Damage or defects affecting function or serviceability	X	
Damage or defects not affecting function or serviceability		X

4.4 Test. Test items selected in accordance with 4.2.2 as specified in 4.5. Lot acceptance shall be based on an AQL of 4 percent defective.

4.5 Test method.

4.5.1 Plastic parts. Test plastic parts for compliance to 3.3.1 in accordance with ASTM D570. Failure to pass this test shall cause rejection of the lot.

4.5.2 Chrome plating thickness. Test thickness of plating in accordance with non-destructive tests in QQ-C-320.

4.6 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

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TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing, damaged or not conforming to the applicable specification or requirements stated herein.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling or distortion of container.
Contents	Quantity per container is more or less than required. Gross weight exceeds requirements of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Twelve rods of like description or twelve flush bulbs of like description or one flush valve or one lever assembly shall be packaged in a close-fitting box conforming to PPP-B-636, class weather resistant, style optional. Twelve unit packages of rods (144 rods) or 25 unit packages of lever assemblies (25 lever assemblies) shall be further packaged in a close-fitting box conforming to PPP-B-636, class weather resistant. Close and waterproof seal the box in accordance with the appendix to PPP-B-636.

5.1.2 Level B. Twelve rods of like description or twelve flush bulbs of like description or one flush valve or one lever assembly shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676. Twelve unit packages of rods (144 rods) or 25 unit packages of lever assemblies (25 lever assemblies) shall be further packaged in a close-fitting box conforming to PPP-B-636, class domestic. Closure of boxes shall be in accordance with the applicable box specification or appendix thereto.

5.1.3 Level C. One-hundred and forty-four rods of like description or twelve flush bulbs of like description or one flush valve or 25 lever assemblies of like description, shall be packaged to insure protection during shipment and safe delivery to destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve packages of rods (1728 rods) or twelve packages of flush bulbs (144 bulbs) or thirty-six packages of flush valves (36 valves) or four packages of lever assemblies (100 lever assemblies) of like description packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class weather resistant. Close, waterproof seal and strap the boxes in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Twelve packages of rods (1728 rods) or twelve packages of flush bulbs (144 bulbs) or thirty-six packages of flush valves (36 valves) or four packages of lever assemblies (100 lever assemblies) of like description packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. Closure of the boxes shall be in accordance with Method II of the appendix to PPP-B-636.

5.2.3 Level C. Rods, bulbs, valves or lever assemblies packages as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Float valves, float valve bulbs, and trip lever assemblies covered by this specification are intended for use in water closet tanks.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Definitive part number (see 1.2).
(c) Level packaging and packing required (see 5.1 and 5.2).
(d) When special markings are required (see 5.3).

6.3 Definitive part number, example. The Federal specification number and the specification part number combine to form the definitive part number.

Example:

DPN ---------- WWV1916 - 01
Fed. Spec. ----------
Flush valve (see 1.2.2) ----------

6.4 Illustrations. The illustrations contained herein are for the convenience of identification. They are not intended to preclude the purchase of flush valves of similar design which are otherwise in accordance with the requirements of this specification.

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